

ARSHAD

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PROFESSIONAL SUMMARY

Backend Development student (3rd year, TISS) seeking an SDE/Backend Engineering internship. Built and deployed two production systems end-to-end: a Spring Boot/WebSocket messaging application (MongoDB, Docker, CI/CD to Render) and a PHP/MariaDB Hospital Management System (deployed on InfinityFree). Comfortable across REST API design, event-driven backend logic, and both relational and NoSQL databases.

EDUCATION

Bachelor of Software Development | Tata Institute of Social Sciences (TISS) Expected 2027

Mumbai, India | Current GPA: 8.4/10 | 2nd Year

- Relevant coursework: Data Structures & Algorithms, OOP, Database Management Systems, Software Engineering.

KEY ACHIEVEMENTS

- Independently built, containerized, and deployed a real-time messaging application to production (Spring Boot, WebSockets, MongoDB, Docker, Render CI/CD).
- Diagnosed and fixed a privilege-escalation vulnerability, broken JOIN queries, and a failing password-hashing flow in a live PHP/MariaDB system, then deployed it publicly on InfinityFree.

TECHNICAL SKILLS

- **Languages:** Java, PHP, JavaScript, C, HTML5, CSS3
- **Frameworks:** Spring Boot, Spring Security, Spring MVC, RESTful API Design, Native WebSockets
- **Databases:** MongoDB (NoSQL), MySQL, MariaDB (Relational), Single Collection Design patterns
- **Architecture:** Event-Driven Architecture (ApplicationEventPublisher), Concurrent Programming, OOP, Design Patterns (Repository, Factory, Singleton)
- **Testing:** JUnit (unit testing), integration testing, Postman (API testing)
- **Tools:** Git, GitHub, Docker, Maven, Gradle, Ubuntu Linux (CLI), XAMPP, VS Code
- **Concepts:** Data Structures & Algorithms, System Design, CI/CD pipelines, Agile/Scrum methodology

PROJECTS

Real-Time Enterprise Messaging Engine (Deployed) | Web Application Java, Spring Boot, WebSockets, MongoDB, Docker

Live Demo: chat-application-hvp1.onrender.com

- Architected a real-time messaging backend by migrating from STOMP to Native WebSockets, reducing per-message protocol overhead and simplifying the connection lifecycle for standard text frames.
- Engineered a constant-time message routing system using a thread-safe ConcurrentHashMap to track active user connections and route messages to the correct session.
- Implemented Spring Security authentication by binding JSESSIONID at the HTTP handshake level, preventing identity spoofing on WebSocket connections.
- Built a decoupled, event-driven architecture using Spring's ApplicationEventPublisher to move background processing (e.g. room cleanup, notifications) off the main request thread.
- Designed a unified MongoDB schema (single-collection design) to support both 1-on-1 direct messages and group chats with consistent query patterns.

Hospital Management System (Deployed) | Web Application PHP, MariaDB, HTML5, CSS3, JavaScript

Live Demo: hospital-management-system-arshad.gt.tc **GitHub:** [kentakaa/hospital-management-system](https://github.com/kentakaa/hospital-management-system)

- Debugged and stabilized an MVC-restructured Hospital Management System, fixing routing errors, a privilege-escalation vulnerability, and broken JOIN queries across patient, doctor, and ward modules.
- Fixed a broken bcrypt password-hashing flow and rebuilt 22 missing view files by recovering and reconstructing logic from Git history, restoring core application functionality.
- Tested the full application locally using XAMPP, resolving remaining functional and UI bugs before deploying the final build publicly on InfinityFree.

WORK EXPERIENCE

Software Development Intern | PC Training Institute (PCTIL) Aug 2025 – Oct 2025

- Built the billing and appointment-scheduling modules of a hospital management system backend in PHP and MySQL, working within an existing multi-module codebase.
- Optimized SQL queries and added indexing on frequently-filtered columns to speed up appointment and billing record lookups.
- Built a patient onboarding form with server-side validation to reduce manual data-entry errors during scheduling.

CERTIFICATIONS & CONTINUOUS LEARNING

Software Development Job Simulation | Citi ICG Technology (via Forage) April 2026

- Built a concurrent Java data-ingestion pipeline for streaming price updates, and a real-time JavaFX stock tracker integrated with the Yahoo Finance API using Google Guava caching.